

ANTIFUNGAL DRUGS

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OVERVIEW

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- ▶ Fungal infections are divided into three groups namely; systemic, subcutaneous and superficial mycoses
- ▶ **Systemic infections** are mostly caused by *Aspergillus*, *Blastomycocetes*, *Candida*, *Coccidioides*, *Cryptococcus* and *Histoplasma* the diseases of which tend to be invasive and life threatening
- ▶ *Aspergillosis*, *candidiasis*, *cryptococcosis* and *mucomycosis* are more likely to occur in immunocompromised individuals
- ▶ **Subcutaneous mycoses** are mostly caused by puncture wounds contaminated with soil fungi
- ▶ **Superficial mycoses** are caused by yeasts and dermatophytes which affects the nails, skin and mucous membranes

Further discussion on superficial mycoses

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Dermatophytes are named differently depending on their location i.e,

- ▶ Infections of the nails – Tinea unguium or onychomycosis
- ▶ Infection of the feet – Tinea pedis or Athletes foot
- ▶ Rings worms of the body – Tinea corporis
- ▶ Rings worms of the head – Tinea Capitis
- ▶ Infections of the inguinal, genitals, buttocks – Tinea cruris (Jock itch)

- ▶ All these infection manifest with pruritus (itching) and erytherma

- ▶ Ring worms present as annular (ring shaped) scaling rash with a clear center

- ▶ Most yeast infections are caused by candida albicans and other candida types and those affected may have thrush (Oral candidiasis), vaginal candidiasis, candida infections of the axilla, groin and gluteal folds including diaper rash in children

- ▶ Pityriasis versicolor or tinea versicolor, a skin condition characterized by hypopigmented and hyperpigmented macules presentation mostly in the shoulder girdle is caused by *Malasezia furfur*(also called *Pityrosporum ovale*)

Fungal cell membrane, the target for many antifungal drugs

- ▶ Fungal cell membranes contain ergosterol while the mammalian cell membranes contain cholesterol
- ▶ This difference in the two sterols forms the basis for the selective toxicity of antifungal drugs
- ▶ Some antifungal drugs bind to ergosterol thereby, increasing the plasma membrane permeability
- ▶ Using different mechanisms, another set of drugs inhibit the synthesis of ergosterol
- ▶ Tempering with the integrity of the cell membrane or inhibition of its synthesis whether directly or indirectly accounts for the mechanism of action for larger part of antifungal drugs

Classification of antifungal drugs

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POLYENES	ALLYLAMINES	ECHINOCANDINS
Amphotericin B Natamycin Nystatin	Terbinafine Naftifine	Caspofungin
AZOLES DERIVATIVES		OTHERS
IMIDAZOLES (DIAZOLES)	TRIAZOLES	
Clotrimazole Econazole Ketoconazole Miconazole Tioconazole	Fluconazole Itraconazole Voriconazole Posaconazole	Griseofulvin Flucytosine Tolnaftate Ciclopirox

NB: The Azole derivatives contain a five member azole ring, with the imidazole(diazole) group having two nitrogens while the triazoles have three nitrogens

POLYENES

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- ▶ Amphotericin B is the only polyene drug used for systemic and subcutaneous mycoses
- ▶ Nystatin and natamycin are limited to topical preparations for treatment of superficial mycoses of the skin and the mucous membranes

Mechanism of action

- ▶ Polyenes selectively bind to ergosterol, an important component of fungal cell membrane
- ▶ This leads to increased permeability in the plasma membrane thus leading to leakage of the cytoplasmic content out of the cell

Amphotericin B

- ▶ The drug is not absorbed from the gut and is thus available as a parenteral preparation
- ▶ It is also available in topical preparations for the treatment of superficial infections
- ▶ The dose and route of administration is dependent on the site and severity of the infection
- ▶ Higher doses are used to treat infections caused by resistant fungi like *Aspergillus* species while lower doses are used to treat esophageal and urinary tract infections

Indications

- ▶ Active against a wide variety of fungi infections including cryptococcal meningitis, histoplasmosis

Adverse effects

- ▶ Amphotericin B has been called “**Ampho - terrible**” as it is believed to be the most toxic antimicrobial drug in use today causing the following;
- ▶ Nausea, vomiting, diarrhea (When given parenterally)
- ▶ Renal toxicity leading to hypokalemia and hypomagnesemia
- ▶ Cardiotoxicity which includes arrhythmias and BP changes
- ▶ Neurological disorders including hearing loss, convulsions, diplopia, peripheral neuropathy, encephalopathy
- ▶ Liver dysfunction
- ▶ Rash
- ▶ Thrombophlebitis at the site of injection

Lipid formulations of Amphotericin B

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- ▶ Apart from the traditional deoxycolate formulation amphotericin B, there are other lipid formulation i.e.
 1. Amphotericin cholesteryl sulphate (AMPHOTEC)
 2. Amphotericin phospholipid complex (ABELCET)
 3. Amphotericin liposomal complex (AMBISOME)
- ▶ These formulations have better pharmacokinetic properties that reduce renal concentration and toxicity and hence they are preferred in persons with renal impairment or those intolerant to the traditional formulation
- ▶ After administration of lipid formulation, the drug is sequestered in the cells of the liver and the spleen which gradually and slowly releases the drug over several days that produce plasma concentration that is adequate enough to actualize treatment

Nystatin and Natamycin

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Nystatin

- ▶ Active against *Candida* species and is available in various topical preparations like creams, ointments and powders for mucocutaneous candidiasis
- ▶ Orally administered tablets and suspensions are available for oral and intestinal candidiasis while vaginal pessaries are available for vaginal candidiasis

Natamycin

- ▶ Active against *Aspergillus*, *Candida*, *Fusarium* and *Penicillium* species
- ▶ Is available as an ophthalmic suspension for the treatment of fungal blepharitis, conjunctivitis or keratitis

AZOLE DERIVATIVES

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- ▶ These are synthetic drugs used in the treatment of various mycoses
- ▶ Some azoles are applied topically to treat superficial infections while others are administered for the systemic infections for more stubborn infections like onychomycosis
- ▶ Several azole derivatives reserved for systemic infections are given orally while fluconazole and voriconazole can also be administered intravenously for more serious infections
- ▶ Absorption of ketoconazole, itraconazole and posaconazole requires the presence of gastric acid and hence drugs that reduce gastric acid should not be administered concurrently
- ▶ Azole derivatives undergo considerable hepatic biotransformation and the parent compound and metabolites are excreted in the urine and stool

Mechanism of Action of Azole derivatives

- ▶ They block the biosynthesis of ergosterol by inhibiting 14 α demethylase, a cytochrome P450 that converts lanosterol to ergosterol
- ▶ When ergosterol synthesis is inhibited then cell plasma membrane formation is subsequently blocked or impeded

Indications

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- ▶ Azole derivatives can either be fungistatic or fungicidal depending on the particular organism and drug concentration
- ▶ They are active against a wide range of fungi and also acts as alternatives to amphotericin B for treatment of many systemic and subcutaneous infections
- ▶ Itraconazole is particularly useful in treatment of blastomycosis and histoplasmosis and generally used for onychomycosis (nail infections)
- ▶ Fluconazole achieves better cerebrospinal spinal fluid concentration and is thus used in treatment and prevention of cryptococcal meningitis in patients who have been successfully treated with Amphotericin B
- ▶ Fluconazole is also used to treat mucocutaneous infections like oral pharyngeal and esophageal and disseminated candidiasis like endocardial candidiasis while a single dose has also been proved to be useful in eradicating vaginal candidiasis

Indications of Azoles Cont'd

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- ▶ Voriconazole and posaconazole are referred to as second generation triazoles because of their enhanced activity against *Aspergillus* and *Albicans* species with voriconazole being 60 to 100 times more potent than fluconazole and hence its effectiveness against non albicans like *Candida krusei* and *Aspergillus fumigatus*
- ▶ Ketoconazole is available in oral and topical formulations with the oral formulation less widely used than itraconazole and fluconazole (Currently a recalled product – no longer in use) but the topical preparations are useful for treating seborrheic dermatitis
- ▶ Clotrimazole is available in topical formulations and used for candida infections of the mouth, throat, vagina tinea versicolor, tinea pedis and tinea cruris

ALLYLAMINE DRUGS

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- ▶ Examples are Naftifine and Terbinafine
- ▶ They are available both as topical formulations while terbinafine is also available as an oral formulation

Mechanism of Action of Allylamines

- ▶ Inhibit squalene epoxidase, an enzyme which converts squalene to squalene 2,3 – oxide the immediate precursor of lanosterol
- ▶ Where there is no lanosterol, there will be no ergosterol and subsequently no cell plasma membrane formation

Indications

- ▶ Mostly used for treatment of superficial dermatophyte infections though are also fungistatic against candida species
- ▶ Terbinafine is administered orally for treatment of onychomycosis with finger nail infections requiring 6 weeks of therapy while toe nail infections requiring 12 weeks of therapy

Adverse effects

- ▶ Both drugs are well tolerated and rarely cause serious adverse effects

ECHINOCARNDINS DRUGS

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- ▶ Examples are caspofungin, anidulafungin and micafungin

Mecanism of Action of echinocarndin drugs

- ▶ These inhibit the fungal cell wall synthesis by non competitive inhibition of β -(1,3) D – glucan synthatase enzyme complex, an enzyme which enhances formation of one important component of the cell wall called β -(1,3) D – glucan

Indications

- ▶ Caspofungin excellent activity against candida species and good activity against Aspergillus species and is also active against candida strains know to be resistant to azole derivatives
- ▶ Anidulafungin and micafungin are newer drug which are recommended for prophylaxis and treatment of certain candida infections and invasive Aspergillosis

Adverse drug effects

- ▶ Common ones are headache, fever, phlebitis at site of injection and abnormal liver function results
- ▶ Generally and when compared to amphotericin B, echinocandins have few adverse effects

OTHER ANTIFUNGAL DRUGS

FLUCYTOSINE

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Mechanism of Action

- ▶ The drug accumulates in the cell membrane and is converted to its metabolite 5 - Fluouracill (5-FU) by cytosine deaminase
- ▶ This metabolite is then incorporated into the fungal RNA and thus interfering with fungal protein synthesis

Indications

- ▶ Flucytosine is mostly used in combination with amphotericin B because the easiness with which fungal resistance develops resulting from mutations in genes encoding cytosine deaminase, cytosine pernease or enzymes that incorporate 5 – FU into fungal RNA
- ▶ This combination flucytosine and amphotericin B produce a synergistic effect against Candida and Cryptococcus which is effective in the treatment of pneumonia, meningitis, endocarditis or septicemia caused by these organisms

Adverse effects

- ▶ Usually mild though some people experience haematologic toxicity while cardiopulmonary arrest have been reported in some patients

GRISEOFULVIN

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- ▶ It is a fungistatic drug derived from *penicillium griseofulvum*
- ▶ It is active against numerous dermatophytes but not active against candida and other fungi

Mechanism of Action of Griseofulvin

- ▶ Griseofulvin is deposited in keratin precursor cells of the skin, hair and nails where it disrupts microtubule function and hence leading to the inhibition of mitosis of susceptible dermatophytes
- ▶ The infected cells are gradually exfoliated and replaced by non infected tissue

Indications

- ▶ Griseofulvin is the standard treatment of tinea capitis which is mostly caused by *Tinea tonsurans*
- ▶ Some infections respond to griseofulvin in 2 to 8 weeks while persistent infection may require 3 – 6 months

Side effects of griseofulvin

- ▶ Headaches
- ▶ Gastrointestinal upset – nausea, vomiting, heartburn, cramps, flatulence
- ▶ Rashes including urticarial, lupus erythematosus
- ▶ Urinary disturbances: frequency, bed wetting
- ▶ Nervous system disturbance: dizziness, depression, nightmares, fatigue, blurred vision
- ▶ Menstrual disturbance
- ▶ Liver disturbance

Interactions and Cautions

- ▶ Griseofulvin should not be taken in pregnancy or those couples thinking of conceiving
- ▶ Males should not father children within 6 months of treatment as the drug may damage sperm cells
- ▶ Griseofulvin interacts with H₂ receptor antagonists and anti-acids and so these drugs should not be taken for 2 hours after griseofulvin as they disrupt its absorption

Ciclopirox

- ▶ Ciclopirox is a unique pyridine compound that is active against dermatophytes *C.albicans* and *M.furfer*
- ▶ It is applied topically for the treatment of infections caused by these micro-organism and nail laquer solution is used for treatment of mild onychomycosis

Tolnaftate

- ▶ Tolnaftate is a thiocarbamate drug used in the treatment of versicolor and mild dermatophyte infections
- ▶ As with other topical antifungal drugs, tolnaftate is not reliable for treating infections of the scalp or nail bed

END